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Git Branch Merge

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What is Merging in Git?

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Merging in Git means combining the changes from one branch into another.

This is how you bring your work together after working separately on different features or bug fixes.

Common `git merge` Options

- git merge - Merge a branch into your current branch
- git merge --no-ff - Always create a merge commit
- git merge --squash - Combine changes into a single commit
- git merge --abort - Abort a merge in progress

Change Platform:
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Usually, you first switch to the branch you want to merge *into* (often `main` or `master`), then run the merge command with the branch name you want to combine in.

First, we need to change to the master branch:

Example

```
[user@localhost] $ git checkout master
Switched to branch 'master'
```

Now we merge the current branch (master) with emergency-fix:

Example

```
[user@localhost] $ git merge emergency-fix
Updating 09f4acd..dfa79db
Fast-forward
 index.html | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Since the emergency-fix branch came directly from master, and no other changes had been made to master while we were working, Git sees this as a continuation of master.

So it can "Fast-forward", just pointing both master and emergency-fix to the same commit.

Best Practices for Merging Branches

- Always commit or stash your changes before starting a merge.
- Regularly merge from the main branch into your feature branch to minimize conflicts.
- Read and resolve conflicts carefully-don't just accept all changes blindly.

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Practical Examples

- **Abort a merge:** `git merge --abort`
- **Check status during a merge:** `git status`
- **Resolve a conflict and complete the merge:** Edit the conflicted file(s), then `git add file` and `git commit`
- **Fast-forward merge:** Happens when no new commits diverged-Git just moves the branch pointer forward.
- **No-fast-forward merge:** Use `git merge --no-ff branch` to always create a merge commit, preserving branch history.

As master and emergency-fix are essentially the same now, we can delete emergency-fix, as it is no longer needed:

Example

```
[user@localhost] $ git branch -d emergency-fix
Deleted branch emergency-fix (was dfa79db).
```

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Non-Fast-Forward Merge (`git merge --no-ff`)

By default, if your branch can be merged with a fast-forward (no new commits on the base), Git just moves the branch pointer forward.

If you want to always create a merge commit (to keep history clearer), use `git merge --no-ff branchname`.

Example

```
[user@localhost] $ git merge --no-ff feature-branch
Merge made by the 'recursive' strategy.
 index.html | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Squash Merge (`git merge --squash`)

If you want to combine all the changes from a branch into a single commit (instead of keeping every commit), use `git merge --squash branchname`.

This is useful for cleaning up commit history before merging.

Example

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Aborting a Merge (`git merge --abort`)

If you run into trouble during a merge (like a conflict you don't want to resolve), you can cancel the merge and go back to how things were before with `git merge --abort`.

Example

```
[user@localhost] $ git merge --abort
```

What is a Merge Conflict?

A **merge conflict** happens when changes in two branches touch the same part of a file and Git doesn't know which version to keep.

Think of it like two people editing the same sentence in a document in different ways-Git needs your help to decide which version to use.

How to Resolve a Merge Conflict

Git will mark the conflict in your file.

You need to open the file, look for lines like `<<<<<<< HEAD` and `=======`, and decide what the final version should be.

Then, stage and commit your changes.

Troubleshooting & Tips

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- If you're unsure, ask a teammate or look up the error message.

Merge Conflict Example

Now we can move over to hello-world-images from last chapter, and keep working.

Add another image file (img_hello_git.jpg) and change index.html, so it shows it:

Example

```
[user@localhost] $ git checkout hello-world-images
Switched to branch 'hello-world-images'
```

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Hello World!</title>
<link rel="stylesheet" href="bluestyle.css">
</head>
<body>

<h1>Hello world!</h1>
<div><img src="img_hello_world.jpg" alt="Hello World from Space"
style="width:100%;max-width:960px"></div>
<p>This is the first file in my new Git Repo.</p>
<p>A new line in our file!</p>
<div><img src="img_hello_git.jpg" alt="Hello Git" style="width:100%;max-
width:640px"></div>
```

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Now, we are done with our work here and can stage and commit for this branch:

Example

```
[user@localhost] $ git add --all
[user@localhost] $ git commit -m "added new image"
[hello-world-images 1f1584e] added new image
 2 files changed, 1 insertion(+)
 create mode 100644 img_hello_git.jpg
```

We see that index.html has been changed in both branches.

Now we are ready to merge hello-world-images into master.

But what will happen to the changes we recently made in master?

Example

```
[user@localhost] $ git checkout master
[user@localhost] $ git merge hello-world-images
Auto-merging index.html
CONFLICT (content): Merge conflict in index.html
Automatic merge failed; fix conflicts and then commit t
```

The merge failed, as there is conflict between the versions for index.html.

Let us check the status:

Example

```
[user@localhost] $ git status
On branch master
```

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```
Changes to be committed:
        new file:   img_hello_git.jpg
        new file:   img_hello_world.jpg

Unmerged paths:
  (use "git add ..." to mark resolution)
        both modified:   index.html
```

This confirms there is a conflict in index.html, but the image files are ready and staged to be committed.

So we need to fix that conflict. Open the file in our editor:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Hello World!</title>
<link rel="stylesheet" href="bluestyle.css">
</head>
<body>

<h1>Hello world!</h1>
<div><img src="img_hello_world.jpg" alt="Hello World from Space"
style="width:100%;max-width:960px"></div>
<p>This is the first file in my new Git Repo.</p>
<<<<<< HEAD
<p>This line is here to show how merging works.</p>
=======
<p>A new line in our file!</p>
<div><img src="img_hello_git.jpg" alt="Hello Git" style="width:100%;max-
width:640px"></div>
>>>>>>> hello-world-images
```

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We can see the differences between the versions and edit it like we want:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Hello World!</title>
<link rel="stylesheet" href="bluestyle.css">
</head>
<body>

<h1>Hello world!</h1>
<div><img src="img_hello_world.jpg" alt="Hello World from Space"
style="width:100%;max-width:960px"></div>
<p>This is the first file in my new Git Repo.</p>
<p>This line is here to show how merging works.</p>
<div><img src="img_hello_git.jpg" alt="Hello Git" style="width:100%;max-
width:640px"></div>

</body>
</html>
```

Now we can stage index.html and check the status:

Example

```
[user@localhost] $ git add index.html
[user@localhost] $ git status
                   On branch master
                   All conflicts fixed but you are still merging.
                     (use "git commit" to conclude merge)

                   Changes to be committed:
```

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```
[user@localhost] $ git commit -m "merged with hello-world-images after fix
                  [master e0b6038] merged with hello-world-images after f
```

And delete the hello-world-images branch:

Example

```
[user@localhost] $ git branch -d hello-world-images
                  Deleted branch hello-world-images (was 1f1584e).
```

Now you have a better understanding of how branches and merging works.

Time to start working with a remote repository!

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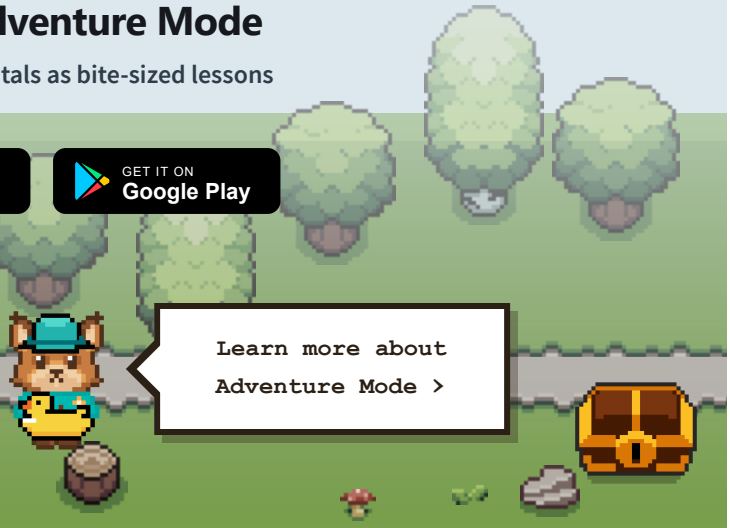
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